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HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING**



CAPSTONE PROJECT

VIDEO QUALITY ENHANCEMENT USING DEEP LEARNING-BASED

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HO CHI MINH CITY
JUNE 2023

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This is to certify that the thesis prepared by Nguyen Quoc Viet and Tran Duc Nam, entitled “**VIDEO QUALITY ENHANCEMENT USING DEEP LEARNING-BASED**” submitted in fulfillment of the requirement for the degree of Computer Science Engineer complies with the regulation of the University and meets the accepted standards with respect to originality.

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Declaration of Authenticity

We confirm that this is our own work, completed under the direction and supervision of Dr. Nguyen Duc Dung and Mr. Bang Ngoc Bao Tam, our findings are authentic and have not been previously reported. We acquired all of the research materials from a variety of sources, and they are all appropriately cited in the references section.

In addition, we included the findings of many other authors and organizations into our research. All of them have been properly referenced.

If plagiarism occurs, we will be held accountable for our actions. As a result, the University of Technology - Vietnam National University HCMC is not responsible for any copyright violations that come from our research.

Ho Chi Minh City, May 2023

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Abstract

Due to scientific and technological advancements, individuals nowadays have much greater expectations for life, especially in terms of entertainment. It is essential to watch or view videos of superior quality. There are essentially two ways to create a high-quality video. To begin with, expensive, cutting-edge technology is utilized to record high-quality video. Second, modern technology have made it possible to restore high-quality video material from low-resolution versions. Despite the fact that the first technique often performs better than the second, the second strategy has lately attracted a great deal of interest, particularly from the computer vision community, owing to its usability, performance, and accessibility. To restore or improve video quality to an acceptable level, a deep learning model trained on feature similarities across video frame sequences may be applied. As a consequence, this model may also aid customers in repairing their low-quality films and afterwards enjoying them in a high-quality environment.

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List of Acronyms

AI	Artificial Intelligence
CNN	Convolutional Neural Network
DCN	Deformable Convolutional Networks
HR	High Resolution
LR	Low Resolution
MISR	Multiple Image Super Resolution
MSE	Mean Square Error
NIQE	Naturalness Image Quality Evaluator
PCD	Pyramid, Cascading and Deformable
PSNR	Peak Signal-to-Noise Ratio
PW	Parallel Warping
SISR	Single Image Super Resolution
SOTA	State-of-the-art
SR	Super Resolution
SSIM	Structure Similarity Index Measure
TMSA	Temporal Mutual Self-Attention
TSA	Temporal and Spatial Attention
VSR	Video Super Resolution

Chapter 1

Introduction

1.1 Problem Statement

In terms of reconstruction accuracy and computing performance for Single Image Super-Resolution, various models based on deep neural networks have achieved remarkable success during the past few years. Before reconstruction, the low-resolution (LR) input picture is upsampled to the high-resolution (HR) space using a single filter, often bicubic interpolation. This indicates that the super-resolution (SR) process occurs in HR space. Due to the lack of matched data, high-resolution photos, and their low-resolution counterparts, researchers must assume that LR data is only a low-pass filtered (blurred), randomly downsampled, and noisy version of HR data. In other words, in order to handle Single Image Super-Resolution (SISR), we must develop a model that can map LR to HR space. Transitioning from photos to videos introduces an extra data dimension (time), a temporal component with a high degree of correlation that may be used to increase both accuracy and efficiency.

The first methods try to map the LR space to the HR space using a single filter, such as bicubic interpolation, before applying the upsampling techniques to get the SR data. These techniques continue to face severe obstacles. First, they are unable to comprehend or fully utilize the relationships between LR and HR. For instance, we downsample the HR several frames (LR data) of a movie using a series of linear and non-linear functions, and then we compel the model to learn how to utilize certain predefined single-filters to rebuild these LR (data) to HR (data). Then, obviously, the model's performance will decline, and this

is where CNN - Convolution Neural Network and DCN [12] - Deformable Convolutional Networks come into action. Second, our topic is Video Super-Resolution (VSR), not Single Image Super-Resolution (SISR) or Multiple Images Super-Resolution (MISR), so it is impractical to use only single filters as the spatial transformer networks, which provide a means to infer parameters for a spatial mapping between two images, or encode optical flow features with unsupervised training. However, they have not yet been examined for video motion correction. Therefore, an additional component, termed the Temporal transformer, is required to use the redundancy properties across frames over an extended period of time.

So, based on these improvements, recent methods entail four interrelated components, namely, **propagation**, **alignment**, **aggregation**, and **upsampling** for VSR problems. In details:

1. **Propagation** refers to the way in which features are propagated temporally.
2. **Alignment** concerns on the spatial transformation applied to misaligned images/features
3. **Aggregation** defines the steps to combine aligned features
4. **Upsampling** describes the method to transform the aggregated features to the final output image.



Figure 1.1: The illustration of super-resolution task [1]

1.2 Thesis goals

This thesis is about building a deep learning Video Super-Resolution model that will take LR videos as input and then transform them into the SR space (x4 upscale). The output of this model must meet three requirements:

1. **Spatial preservation** The features within the frame must be preserved as much as possible.
2. **Temporal leverage** Leveraging the additional temporal dimension to extract temporal features and enhance the output from spatial features.
3. **Computational Efficiency** The deep learning model can be trained and deployed under the barrier of computational budget of undergraduate students.

1.3 Thesis scope

Video Super-Resolution task is a broad topic that has a wide range of applications. In this thesis, we focus on the problem of extend the task of Video Super-Resolution to real-world video - which is captured by multiple type of devices and settings.

Due to the goals of Computational Efficiency define above, we only develop a model that is robust within relative affordable training steps, the model must be light-weight to fit in Google Colab GPU.

Under the time constraint of thesis, we will make some implicit choices in the model that may not be tested while doing our best to explain our tested choices with metric numbers provided in the later chapter of the thesis.

1.4 Contributions

Since we are living in an era of short videos visual technology with Tiktok, Facebook Watch, Instagram Reels, Youtube Shorts... with multiple content makers who make videos with devices ranging from dated Smartphone camera to high-end expensive hardware. These videos often captured in the real-world scenery rather than the studio.

First, our research is an attempt to extend the limited research on the Video Super-Resolution in the real-world. We define our model to be a supplier to the increasing demand in the expansion in the world.

Next, recent researches focus deeply to the large model with complex architecture [13], as undergraduate students, there are only small chance to access the GPUs [14] with high computational power. We then develop an approach to mitigate the gap of this problem, realizing a new model with training strategies that is robust in performance.

Lastly, we proposed a new simple scheme of feature extraction for real-world visual contents together with Repeated Propagation base on the assumption of Markov chain [15] and the human native behaviour of memory recall [16].

1.5 Ethics and Sustainability

Our work benefit mostly on the short video industry and serves as a relevant tool or method to provide better source of visual information.

Some societal considerations can be made from this work as we foresee a practical application. Due to the Covid-19 pandemic, students in Vietnam and all over the world have to choose the method of study online. However, the bandwidth stays a tremendous problem as Governments have to ask the Internet providers to expand it. In rural area or low-income people, either the devices to record video to teach online or the devices to receive and display have the problem of poor resolution and quality. This bring us to the idea of this thesis. To scale a little bit into the global level, bring technology is to bring a new era to people, we recognise that the idea of deep learning for video will benefit everyone but significantly on Africa and the Poles area.

On the sustainability perspective, one of the main societal issues with training deep learning models is that they need enormous amounts of energy resources as they require

significant computational power for training over long periods, eventually contributing to the global warming crisis. Training a low computational solution may contribute to the crisis, however, on a long run, we will reduce tremendous effect aiming to recover our Earth.

1.6 Thesis Structure

The total number of chapters in this thesis proposal is five. The first chapter provides an overview of the problem we are attempting to solve and our mission. The second chapter clarifies the ideas of video deterioration and deep learning utilized in this thesis. Section 3 then discusses efforts linked to the Video-Super-Resolution concept. The next section is the heart of this report. In this chapter, we outline our tested approaches in great detail, compare them, and analyze our present course of action. The last chapter provides a summary of the current state of our thesis and our optimistic future plans.

Chapter 2

Theoretical Background

2.1 Video Theory

2.1.1 Image

A digital image is merely a type of data with a grid-like structure. In other terms, it is a numerical representation of a two-dimensional image composed of pixels grouped in rows and columns. Depending on the sort of digital image, pixel values will fall within a distinct range.

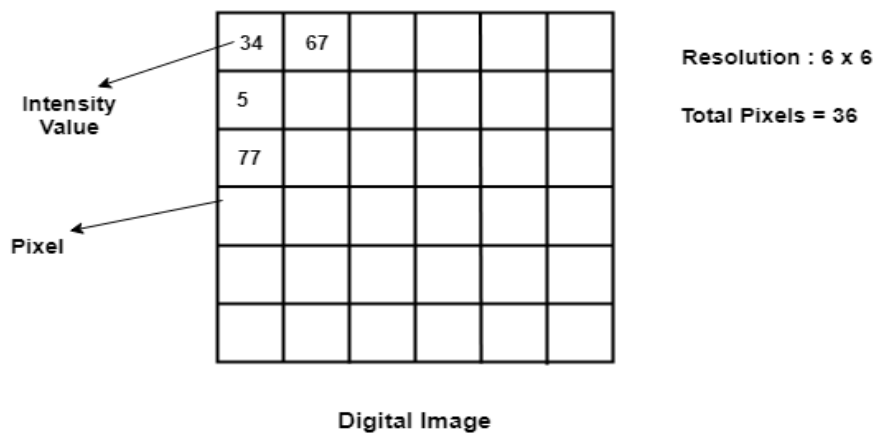


Figure 2.1: An visualization of 8bits 6x6 digital image [2]

2.1.2 Video

This section will describe the fundamental definition and properties of video. This type of definition and characteristics will be crucial information for us to use and build upon in following parts.

A video is a series of images (called frames) that are taken and eventually presented at a predetermined rate. By pausing the sequence at a particular frame, a single video frame, or image, is obtained.

Besides, one of the most important properties of video that also should be recommended is "*redundancy between video frames*". For example:



Figure 2.2: An example of redundancy between video frames [3]

The preceding sample demonstrates a very brief series of frames depicting "Someone is slicing a tomato". Clearly, the content between the frames differs only very slightly (from left to right). Consequently, if we regard each frame as an independent image with no relationship to the others, and construct a deep learning intra-model without employing this redundancy information to upsample the frame sequence, the model will be less accurate. As a result, the intra-output model's will be exceedingly inefficient and time-consuming. Therefore, an inter-model that acknowledges that the sequence of frames has some relationship and attempts to use this information will be more rational and practicable.

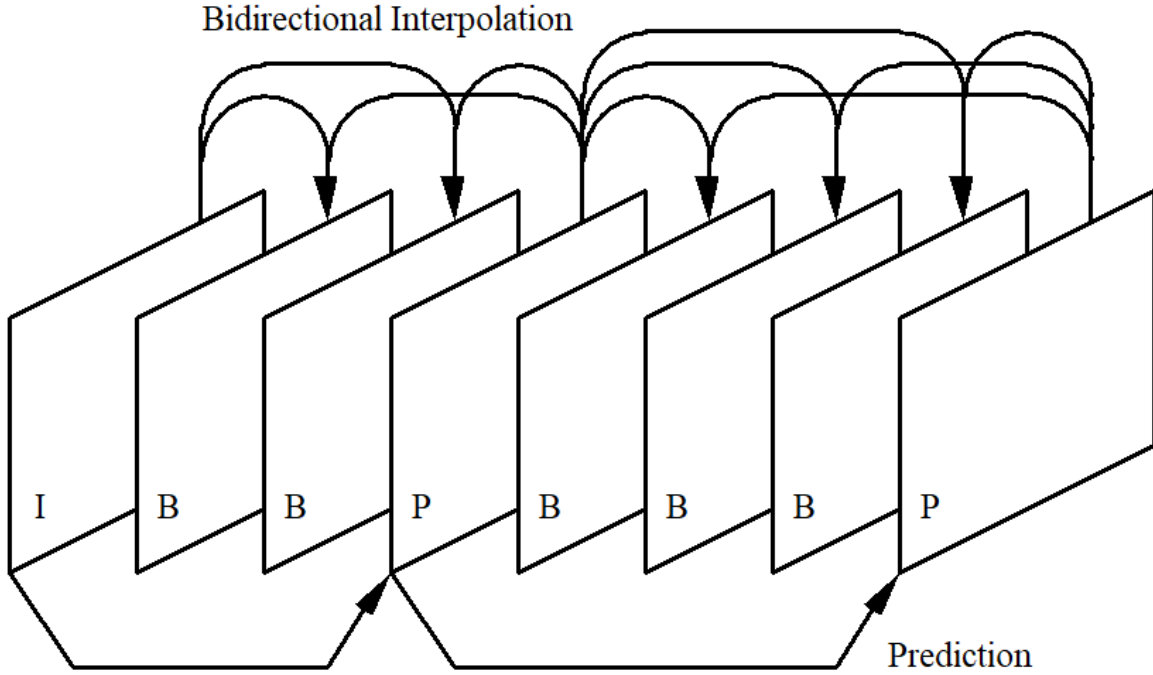


Figure 2.3: Inter model visualization [4]

The first image (I-picture) at the leftmost is upsampled using only the pixels in the image. The fourth one (P-picture) is predictively encoded starting from the I-Picture. The second and third images (B-pictures) are interpolated using the first and the fourth. This continues in the next frames where the sequence can be P-B-B-B-P where the last P-picture is predicted from the first P-picture and 3 interpolated pictures (B-pictures) are created from the first and the last P pictures.

Moreover, another critical property is *frame alignment*. In fact that, not only the objects are moving in video, but the camera also in some cases. Considering this situation, where the camera is dynamic and the objects are static, this will make the frames unaligned with respect to each other. Consequently, the model to build in this project must have an alignment module that can handle the aforementioned problem.

To recapitulate briefly, we present a very basic definition of video. Then based on this definition, we also introduce the two most critical video properties: *redundancy between video frames* and *frame alignment*. For each mentioned property is a task to solve and also its corresponding feasible module.

2.2 Deep Learning theory

2.2.1 Deep Learning

What is Deep Learning? Firstly, Deep Learning is actually a subfield of Artificial Intelligence (AI), particularly Machine Learning.

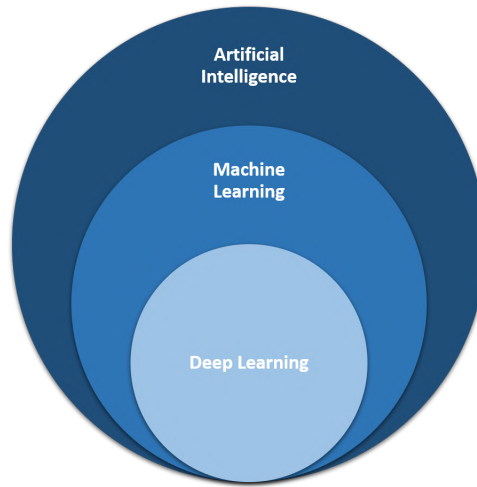


Figure 2.4: Deep Learning scope [5]

Overall, AI is commonly imply a machine or computers that can imitate human intellect and behavior, while machine learning is statistical algorithms that enable AI implementation through data. However, Deep Learning is more meaningful and powerful filed in term of feature extraction and learning these deep feature to exploit the hidden relationship between them by modeling after the human neural brain.

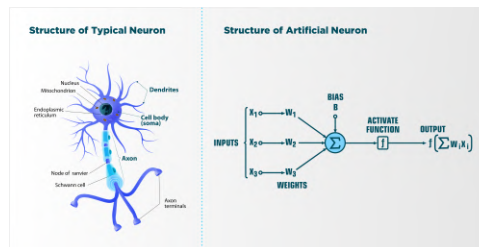


Figure 2.5: Human brain neuron and neural network architectural [6]

In detailed, The brain consists of around 100 billion brain cells – neurons. Each neuron has dendrites (inputs), a nucleus, and axons (outputs). The connection between an axon of one neuron and a dendrite of another neuron is a synapse. The synapse contains a small gap separating the axon and dendrite, and when things are going right, there is dopamine in the synapse that strengthens the electronically connection the two neurons.

In this thesis, we use one class of Deep learning called "Convolutional Neural Network" (CNN) that specializes in processing data that has a grid-like topology, such as an image.

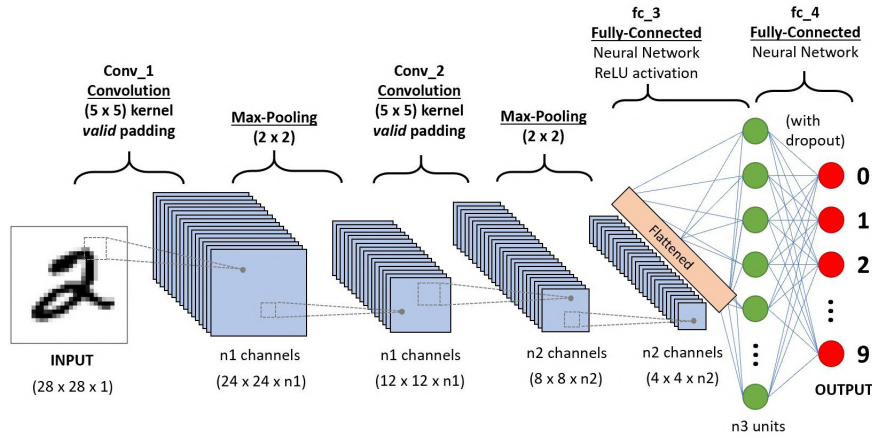


Figure 2.6: Architecture of a CNN [7]

The above figure shows the architecture of a CNN network, which consist three consequently key operations, namely: *convolution*, *activation function*. In some cases, we just need convolution and activation is enough for one step of feature extraction. Later section will also introduce these three key CNN operations.

Conclusion, Deep learning is comparable to programming with data. In the initial configuration of a machine with a neural network, the network is unprogrammed. When pictures are provided to this virtual computer, it begins to program itself to accomplish whatever task is being configured. Deep learning can be utilized in a variety of ways in application fields, particularly image processing, including object recognition, object detection, product rating, image production, etc. In this thesis, we will upsample low-resolution video to high-resolution video using CNN's approach.

2.2.2 Optical flow

Optical flow is the first and the most crucial module or step in all most of video super resolution. It can estimate correctly the motion of objects between consecutive frames of sequence, caused by the relative movement between the object and the camera. The optical flow problem can be illustrated as follow:

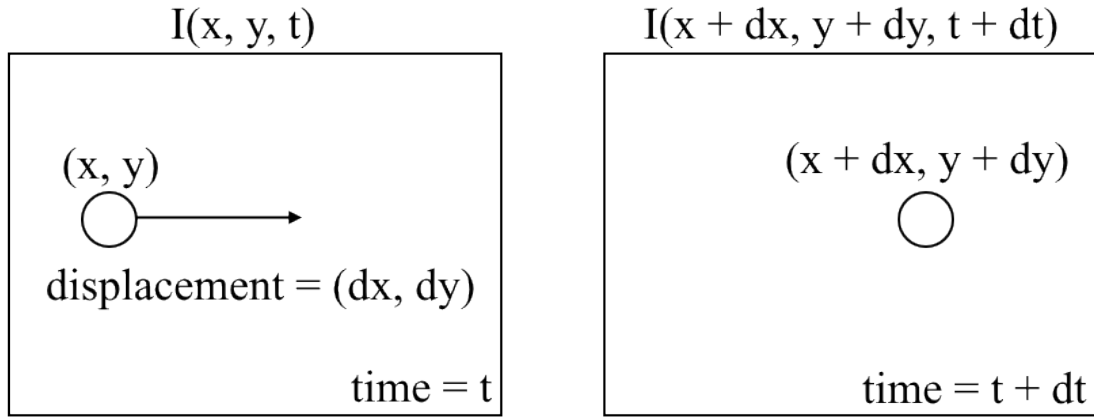


Figure 2.7: Optical flow problem illustration [8]

where between consecutive frames, we can express the image intensity (I) as a function of space (x, y) and time (t). In other words, if we have an image $I(x, y, t)$ and later moving its pixels by a known direction (d_x, d_y along the time dimension t , we can easily obtain the new image $I(x + d_x, y + d_y, t + d_t)$.

In the other word, the optical can represent the flow-field of pixel between consecutive frames. As a sequence, with this flow information we can easily utilize the *redundancy between video frames* since basing on an assumption that "the similar frames will lead to similar flow-field". Another key feature of optical flow is that the optical flow's output can also be used as a feature alignment when combine with one of the most powerful and popular in deep learning - Convolution operation. Deep learning convolution operation will be introduced in the next part.

2.2.3 Convolution

In term of image processing, *Convolution* is the most powerful Deep learning operation to get the deep feature of an image (or frame). Furthermore, Convolution is the core building block of the Convolution Neural Network (CNN).

Convolution works by sliding a window (a kernel matrix) through the input to produce the feature map output (or activation map). The objective of this procedure is to extract the feature that satisfies the kernel constraint. The example of convolution can be presented as follows:

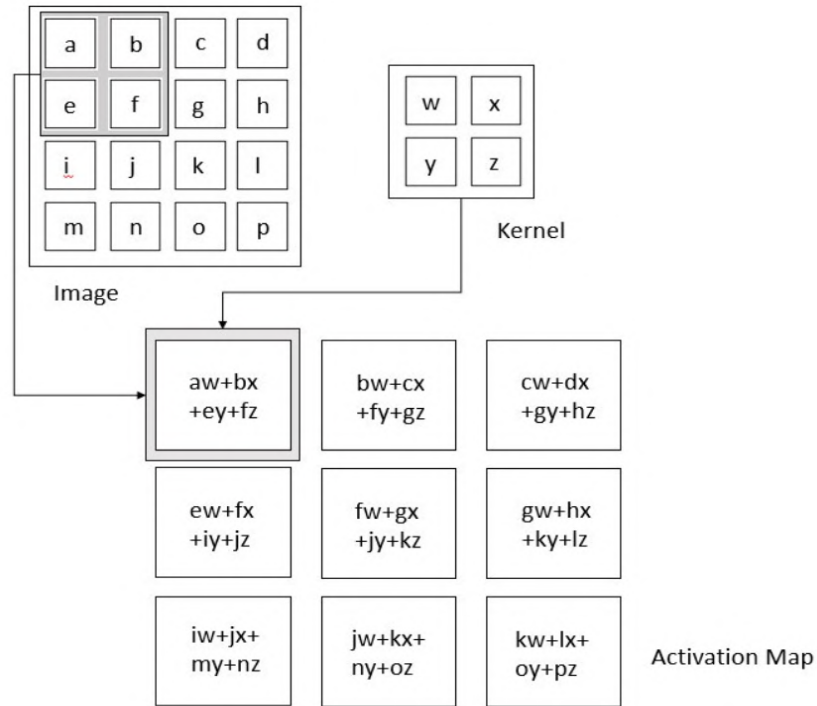


Figure 2.8: Convolution operation [9]

The w, x, y, z in the kernel are learnable parameters, which will be modified via the learning (or training) process of CNN to obtain the most viable parameters that can exploit full feature relationships. And the size of a kernel will be smaller than that of an image, but it will be more comprehensive. This means that if the image consists of three channels (RGB), the kernel height and width will be modest, but the depth will extend to all three channels.

2.2.4 Activation function

For the real world, where most all the objects contact with together by a non-linear relationship. So, capturing this information using the convolution operation, which output the result by just linear combination, is not reasonable enough. Instead, CNN extends a small adjustment step after the convolution, called *activation function*.

The most benefit of activation function is that it *transforms the linear input to non-linear output*. In term of CNN, especially video super resolution, we use *Leaky ReLU* activation function as the most powerful non-linear transformer operation. Leaky ReLU can be described like:

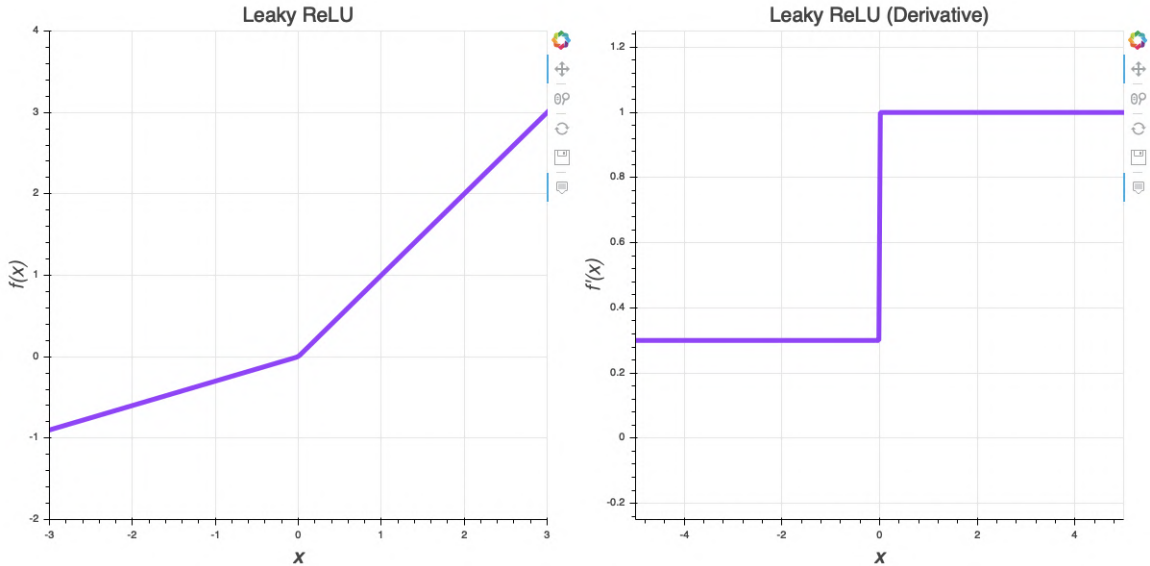


Figure 2.9: Leaky ReLU function

This function will filter all the non-positive and transform to a nearly zero number, since in the image processing all most non-positive values are considered as not important for learning process.

The combination of Convolution and activation in order is not only limited within scope of feature extraction but also can wrap the unalignment feature (alignment step), fusion the propogated features (aggregate step), and even upsampling step in video super resolution task.

2.2.5 Upsampling

Upsampling is the last of the four steps in a standard video super resolution pipe line (propagation, alignment, aggregation, upsampling). Recently, the most popular approach to handle this task is *PixelShuffle* [10].

Pixel Shuffle Super Resolution is an upsampling technique where Image Super Resolution is achieved in a rather ingenious method. Feature maps are extracted in its LR (Low-Resolution) space (as opposed to earlier techniques where this was done in the HR (High-Resolution) space).

The main principal of this method is come from the idea of Efficient Sub-Pixel Convolutional Neural Network (ESPCN [10]).

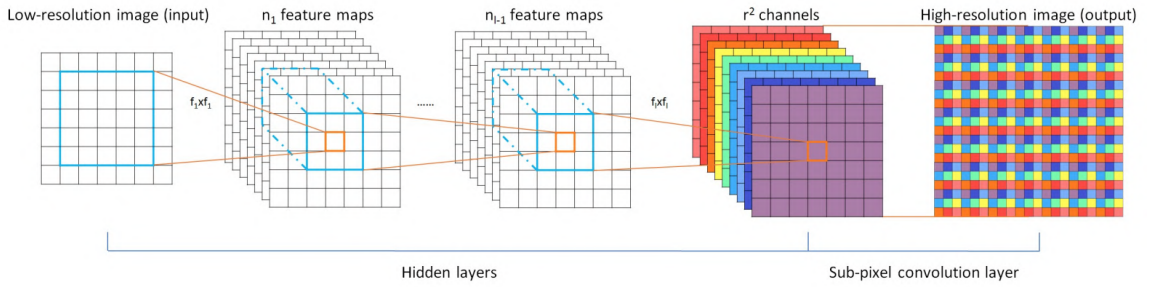


Figure 2.10: Efficient Sub-Pixel Convolutional Neural Network [10]

This approach will first apply a l layer convolutional neural network directly to the LR image (C, H, W) to form an output $(C * r^2, H, W)$, and then apply a sub-pixel convolution layer that upscales this LR feature maps to produce $I^{SR}(C, H * r, W * r)$.

2.3 Evaluation metric theory

In this section, we will introduce three key metrics that are widely used in image quality assessment, namely: *PSNR*, *SSIM*, *NIQE* [17]. But, among those, we can divide into two main type: reference metric (PSNR and SSIM) and non-reference metric (NIQE [17]).

2.3.1 Reference metrics

Reference metrics is the metrics that evaluate the output image base on its corresponding ground truth. For example, we use an extremely advanced camera to capture an image at 1024x1024 resolution, then downsample it by any type of degradation kernels to 256x256 resolution. Then input this 256x256 to an x4 upscaling model, which will give the output at resolution 1024x1024. Finally, compared the output with its corresponding captured 1024x1024 image. In this thesis, we use two reference metrics, namely: *PSNR* and *SSIM*.

Firstly, PSNR stands for "*Peak signal to noise ratio*". It imply the ratio between the maximum possible power of an image and the power of corrupting noise that affects the quality of its representation [def]. PSNR is most easily defined via the mean squared error (MSE) as:

$$MSE = \frac{1}{m \cdot n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2$$

Figure 2.11: Mean squared error

Where m, n is the height and width of both two image. $I(i, j)$ imply the value of a pixel at location (i, j) . K stand for the generated image. After that, the PSNR (in dB) can be computed by formula:

$$\begin{aligned} PSNR &= 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \\ &= 20 \cdot \log_{10} \left(\frac{MAX_I}{\sqrt{MSE}} \right) \\ &= 20 \cdot \log_{10}(MAX_I) - 10 \cdot \log_{10}(MSE) \end{aligned}$$

Figure 2.12: Peak signal to noise ratio - PSNR

Where MAX_i is the maximum possible pixel value of the image. When the pixels are represented using 8 bits per sample, this is 255. PSNR is used to be the most commonly measure the quality of reconstruction of lossy compression. It is an approximation to human perception of reconstruction quality. However, since PSNR is mostly depend on the values of pixel loss (MSE) which can not reflect the similarity in structure between two images. So, it's necessary to consider a evaluation metric that can measure the structural distortion.

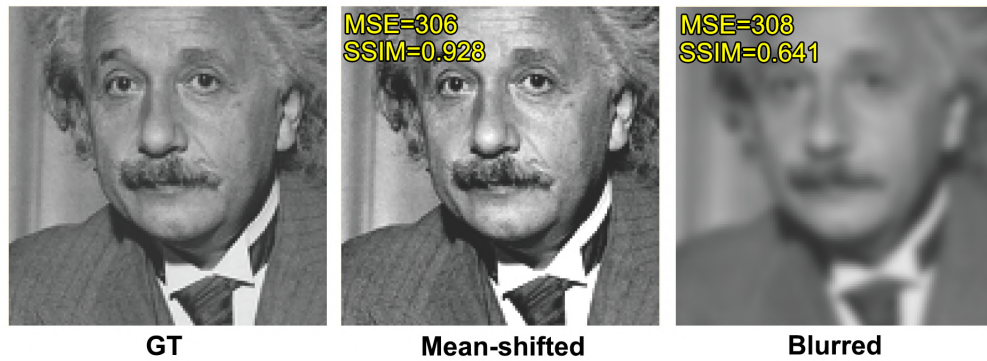


Figure 2.13: Comparison of "Einstein" images with different types of distortions and almost the same MSE [11]

Secondly, SSIM stands for *Structural Similarity Index*. While the MSE can not realize the difference in structure between two images, the SSIM can do it. Since, despite of the fact that calculating the difference by pixel loss, SSIM calculates the difference based on the distribution (mean and variance) of each image. The formula can be described as:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

Figure 2.14: Structural Similarity Index - SSIM

Here,

- μ_x the pixel sample mean of image x ;
- μ_y the pixel sample mean of image y ;
- σ_x^2 the variance of image x ;
- σ_y^2 the variance of image y ;
- σ_{xy} the covariance of image x and y ;
- $c_1 = (k_1L)^2$, $c_2 = (k_2L)^2$ two variables to stabilize the division with weak denominator
- L the dynamic range of the pixel-values (typically this is $2^{Noofbitsperpixel} - 1$);
- $k_1 = 0.01$ and $k_3 = 0.03$ by default.

The range value of SSIM output will be from 0 to 1. The smaller the value to zero, the more different the two images and reverse.

Nowadays, within our scope of video super-resolution, the best approaches [18, 19, 20, 13, 21] use these two metrics to evaluate their output in some testing datasets. Furthermore, the PSNR and SSIM range values, that these approaches can reach, fall between (27-31.5) and (0.82-0.87).

2.3.2 Non-reference metric

Numerous super-resolution algorithms have been proposed in the literature, and most of them focus only on synthetic degradation such as bicubic downsampling and hence they can use reference metrics as a way to evaluate output. But the effectiveness is not clear and the required ground-truth images are not always available in practice. It's necessary to have a non-reference metric that can measure the datasets which has no ground truth but very "real world". That's where NIQE [17] comes to play.

NIQE stands for "Naturalness Image Quality Evaluator" imply the natural of the image. The lower the NIQE value, the better. A NIQE value of 0 is the best.

2.4 Batch length and Sequence length tradeoff

In term of Video super-resolution, we have an extra parameters that need to care which is sequence length implying the length of a sequence frame that model takes for each time training. As a result, with a fixed computational budget, the increased batch size will lead to a decrease in sequence length. That lead us to two cases:

1. Case 1: Training with a larger batch size make the model to perceive more degradations and sence content in each iteration, leading to more stable gradient update.
2. Case 2: Training with a larger sequence length lead the network gain more long-term information for each iteration.

However, one must choose between them when computational is limited. RealBasicVSR [20] are interested in the tradeoff between them with an aim to search for a more effective setting so they do several experiment to get the final conclusion that: "networks trained with longer sequences rather than larger batches could more effectively employ the long-term information in the input sequence, improving stability".

Chapter 3

Related Works

Sliding-window Networks. Some early VSR methods [10] directly use the SISR architectures to reconstruct the video frame by frame. The researcher aims to use consecutive convolutions as a method to estimate directly the optical flow between the fusion of frames, and perform sub pixel (pixelshuffle) to mapping from LR to HR. However, Without proper alignment, the propagated features are not spatially aligned with the input image. Consequently, the information on optical flow can not be aggregation in the proper location, which leads to substandard performance. Later, the product. Later approaches resort to a more sophisticated approach of implicit alignment. An illustration of this is that EDVR [21], introducing the PCD (Pyramid, Cascading, and Deformable) alignment module in which frame alignment is done at the feature level using deformable convolutions in a coarse-to-fine manner.

Recurrent networks. Recurrent framework, in contrast to sliding-window framework, propagates the underlying information throughout the video sequence, demonstrating more powerful potential. For example, BasicVSR family approaches [18, 19, 20] rethinks some components of previous methods and proposes a framework that bidirectional propagation coupled with a simple optical flow-based feature alignment suffices to outperform many state-of-the-art methods. Moreover, the BasicVSR++ [19] and PP-MSVR [22] enhanced the bidirectional propagation to grid propagation which also lead to a new bench-mark of the VSR field.

Multi-Stage Approaches. Unlike SISR methods which often divide the restoration task into smaller easier sub-tasks by sub modules. VSR often contains more complex components. How to integrate the features obtained by different components in the network at various stages is critical to the final recovery result. In EDVR [21], researcher train the two modules: PCA align (Pyramid, Cascading and Deformable Convolution) and TSA fusion (Temporal and Spatial Attention) are trained separately. Although the effect is improved in the end, this non-end-to-end approach is cumbersome and not practical enough.

Alignment For any kind of approach that we mentioned above, whether sliding-window or recurrent approach, have proved that the alignment of video frames is extremely critical for VSR restore tasks. We can divide the alignment methods into two: motion estimation and motion compensation. For the prior, BasicVSR [18] performs alignment at the image level using Spynet [23]. For the later, EDVR use DCN to perform alignment at the feature level. Then the author of BasicVSR++ [19] exploit the main principle behind DCN alignment and flowbased alignment, BasicVSR++ proposes a flow-guided deformable alignment module, which combines the idea of two techniques demonstrated great power.

Fusion. After the features have been extracted from temporal and spatial modules, it is necessary to have an effective fusion methods that can improve the temporal feature fusion. Some early methods use concatenation and convolutions to fuse all feattures from the previous module, which result in feature of adjacent video frames is not fully exploited. EDVR proposes a temporal and spatial attention (TSA) fusion module to integrate features from all aligned frames for subsequent restoration. BasicVSR uses bidirectional propagation to gradually fuse aligned feature. BasicVSR++ uses more aggressive bidirectional propagation arranged in a grid-like manner to repeatedly refine the features, improving expressiveness.

Chapter 4

Our Work

Real-world Video Super-Resolution involves 2 problems: First, Video Super-Resolution related to a long and complex processing pipeline since it needs to aggregate information from not only the spatial dimension but also the temporal dimension. Second, in the real world, almost all datasets for Image/Video Super-Resolution are captured with high frequency devices, resulting in truly perfect representation of distribution in datasets without noise, artifacts, etc., but in this era of electronic devices, more and more camera embedded devices have come into existence, such as camera phones, which face the trade off of low power source and high frequency camera sensor. Low power source hardware frequently results in unclear degradations in image/video produced; additionally, high frequency camera sensors can suffer in varied capturing conditions and, without a doubt, return a noisy output.

4.1 Basic components of Video Super-Resolution

4.1.1 Propagation

One of the most remarkable aspects of VSR is propagation because of the truly nature of how video is represented. Video consists of multiple frames, that leads to the fact that, what extends from the image itself is the temporal dimension. If we choose to ignore this important factor, the VSR task will become the task of Single Image Super-Resolution, by the idea, refuse to extract temporal perspective and may result in many problems, motion inconsistency stands for an example. In this thesis, we will focus on the idea of considering the propagation technique to fully leverage the information from frames because propagation is the key which describes how to make use of the information in a video sequence.

Three well-known methods of propagation

- **Unidirectional Propagation.** In the BasicVSR test example, they use Unidirectional Propagation to further illustrate their insights; this method of propagation allows information to be sequentially propagated from the first frame to the last frame based on the idea of a recurrent model; however, in this setting, the information received by different frames is imbalanced. The first frame, in particular, receives no information from the video sequence other than itself, but the last frame receives information from the entire series. As a result, poor outcomes are to be expected for the early frames. Given an LR image x_i , its neighboring frame x_{i-1} , and the corresponding features propagated from its neighbors, denoted as f_{i-1}^f , we will have:

$$f_i^f = P(x_i, x_{i-1}, f_{i-1}^f) \quad (4.1)$$

where P_b and P_f denote the backward and forward propagation branches, respectively.

- **Bidirectional Propagation.** In BasicVSR and RealBasicVSR [18, 20], this method of propagation is employed, where the information is sequentially propagated from the first frame to the last frame and then another pass captures the flow from the last to the first frame, owing to the simplicity of the implementation and how this

idea had dominated in the realm of Natural Language Processing [24, 25]. Given an LR image x_i , its neighboring frames x_{i-1} and x_{i+1} , and the corresponding features propagated from its neighbors, denoted as f_{i-1}^f and f_{i+1}^b , we will have:

$$\begin{aligned} f_i^b &= P_b(x_i, x_{i+1}, f_{i+1}^b) \\ f_i^f &= P_f(x_i, x_{i-1}, f_{i-1}^f) \end{aligned} \quad (4.2)$$

where P_b and P_f denote the backward and forward propagation branches, respectively.

- **Coupled Propagation.** In IconVSR [18], they extend the equation of Bidirectional propagation by a concatenation at the input to permit connection between branches.

$$\begin{aligned} f_i^b &= P_b(x_i, x_{i+1}, f_{i+1}^b) \\ f_i^f &= P_f(x_i, x_{i-1}, f_{i-1}^f, f_i^b) \end{aligned} \quad (4.3)$$

where P_b and P_f denote the backward and forward propagation branches, respectively.

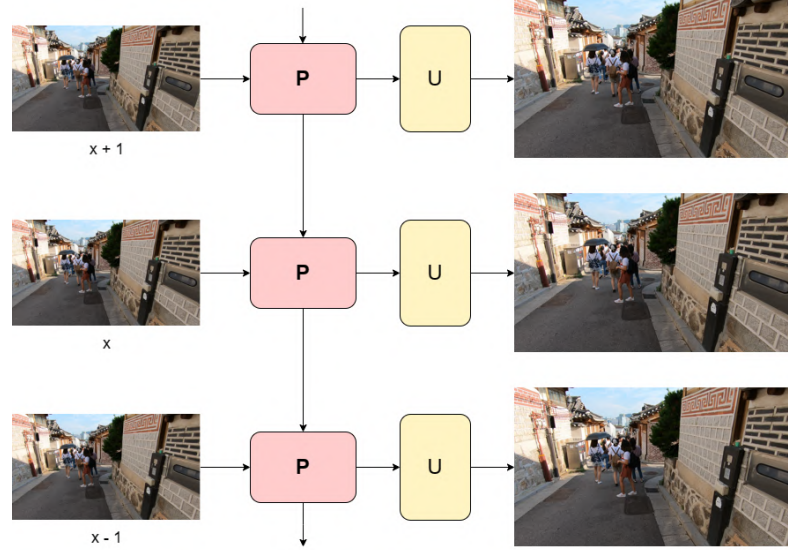


Figure 4.1: Unidirectional propagation - BasicVSR method

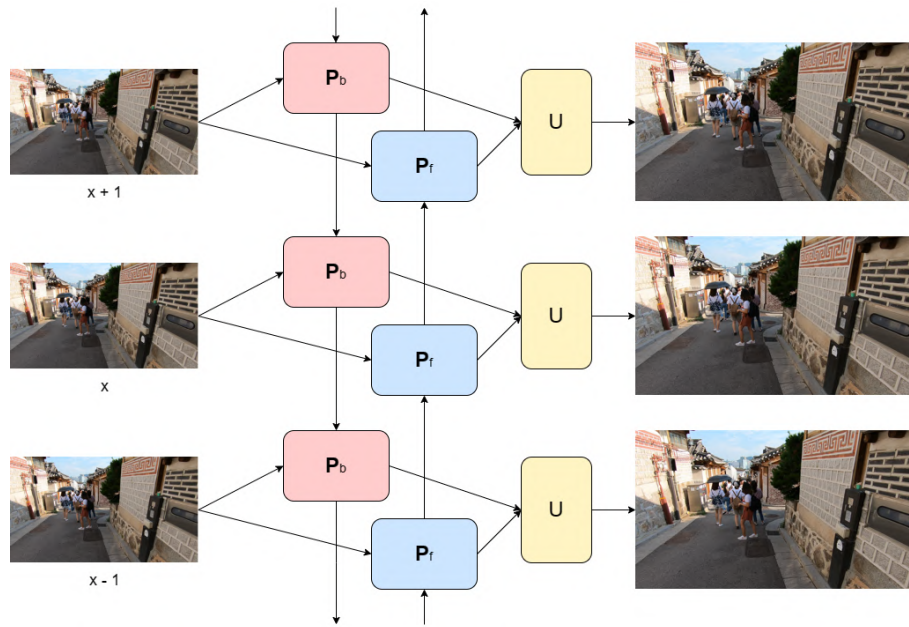


Figure 4.2: Bidirectional propagation - BasicVSR architecture

Comparison of above propagation methods

Unidirectional vs Bidirectional propagation

Early timesteps get less information in unidirectional propagation, resulting in poor per-

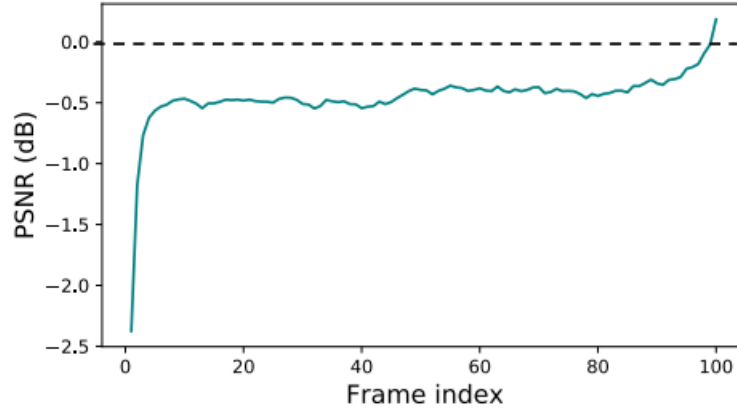


Figure 4.3: Unidirectional vs Bidirectional propagation

formance. Values less than zero (dotted line) imply that the PSNR is lower than the bidirectional equivalent. Due to the bidirectional model's zero feature initialization, the unidirectional model outperforms the bidirectional model only on the final frame.

	BasicVSR	IconVSR (w/o refill)	IconVSR
Info-Refill	✗	✗	✓
Coupled-Prop	✗	✓	✓
PSNR	31.42/30.17	31.60/30.38	31.67/30.45

Figure 4.4: Bidirectional vs Coupled propagation

Bidirectional vs Coupled propagation

The above image 4.4 is of the Evaluations of IconVSR components. The IconVSR model without information refill (middle) is nothing but the model of BasicVSR but with the connection permitted. As observe, the PSNR is raised by the value of 0.18.

Observation: More propagation branches and connections give better output results.

Inspect the three propagation strategies, we retrieved some basic insights on the logic how they seem to work well.

Unidirectional propagation is a simpler and more basic strategy since it only requires information from one direction to be considered. This method can work effectively in videos with generally solid and consistent material between frames. However, unidirectional propagation may not capture all of the relevant information required to improve the resolution of the current frame in videos with a lot of motion or changes in content.

By employing both past and future frames, bidirectional propagation can collect more information. This method is especially useful in videos with a lot of motion or changes in content since it may better collect the important information required to increase the resolution of the current frame. Bidirectional propagation, on the other hand, is more computationally demanding and requires more processing power than unidirectional propagation. Bidirectional propagation, however, still being outperformed by the unidirectional propagation because of the zero feature initialization.

Coupled propagation leverages the key idea of knowledge passing from one propagation branches to another. More computational demanding is also observed. However, the PSNR raises show the applicability of the idea.

From one to two propagation branches, we acknowledged that the PSNR is getting higher, from two propagation branches to connected branches we also get that result, the question that we ask ourselves now turn to be: "If we have more propagation branches and permit a connection over all of it, how will it be?"

Repeated Propagation

To answer the question we stated above, we proposed the Repeated propagation technique to permit recurrent refining through propagation, motivated by the efficacy of bidirectional propagation above 4.3. More specifically, the intermediate features are alternately propagated backward and forward in time. Propagation allows information from separate frames to be "recalled" - in the term of human memory processing, from their previous learning propagation flows. We simply make the duplication into 4 branches, allow the forward and backward features to be "repeated" and "recalled" at the third and forth propagation flows while solving the problem of zero feature initialization thank to the mechanism of "repeated" branches. In contrast to previous researches that propagate features only once, Repeated Propagation collects information from the full sequence many times, accumulates it, boosting feature expressiveness while still remain the assumption of Markov property. Expressed in the formula, given an LR image x_i , its corresponding recall features rf_i , propagation features pf_{i-1} and pf_{i+1} , its neighboring frames x_{i-1} and x_{i+1} , we have:

$$\begin{aligned} f_i^b &= P_b(x_i, x_{i+1}, rf_i, pf_{i+1}) \\ f_i^f &= P_f(x_i, x_{i-1}, rf_i, pf_{i-1}) \end{aligned} \quad (4.4)$$

where P_b in $\{P_{b1}, P_{b2}\}$ denote the first and second backward propagation branches, P_f in $\{P_{f1}, P_{f2}\}$ denote the first and second forward propagation branches. rf_i is defined the to be the dynamic recall feature.

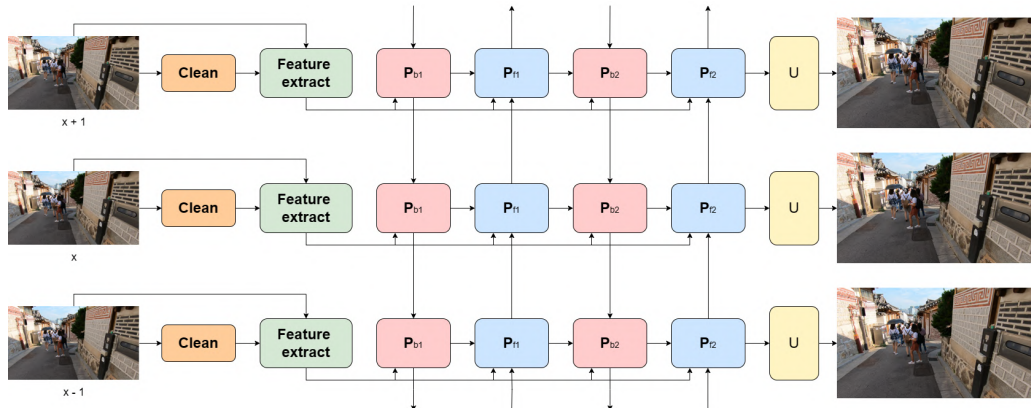


Figure 4.5: Repeated propagation - Our model architecture

Repeated propagation vs Coupled propagation

We investigate the effectiveness of our proposed propagation scheme compare to the Coupled propagation of IconVSR by doing experiment. We calculate the NIQE and PSNR score at checkpoint number 25000, 50000 and 75000 of phase 1 with the same training setting. We use REDS4 validation set to examine the scores.

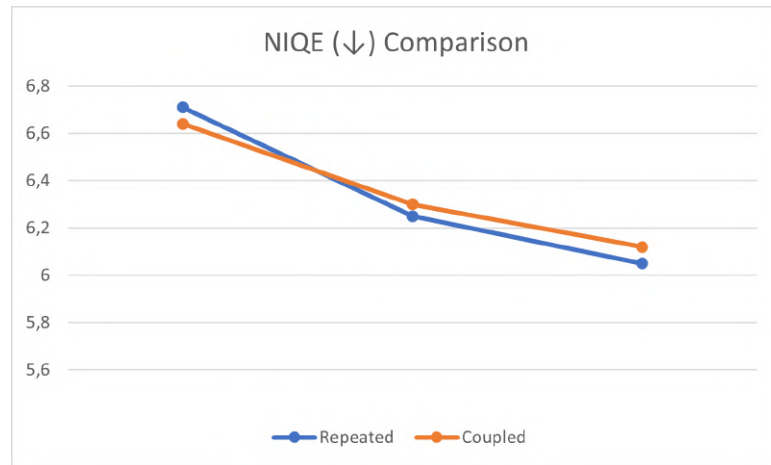


Figure 4.6: Repeated vs Coupled propagation - NIQE

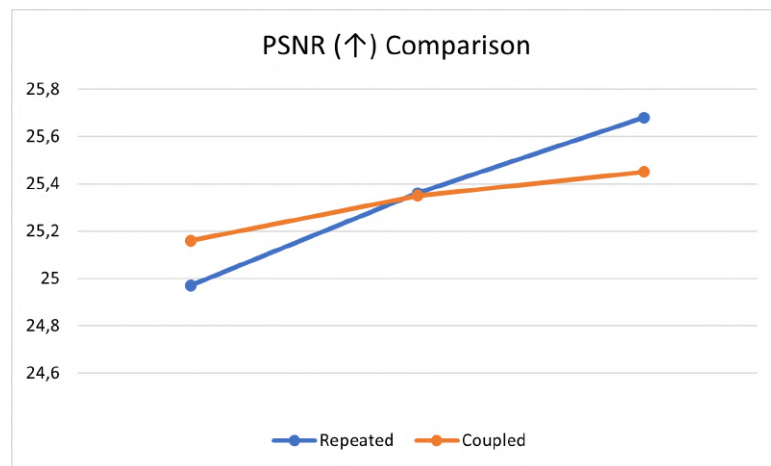


Figure 4.7: Repeated vs Coupled propagation - PSNR

We observe that, at the checkpoint 25000, NIQE and PSNR of Coupled propagation shows dominant in the score, however with the checkpoint of 50000, the two method of propagation give the score to be close to each others, the better belong to Repeated. Have been trained to the checkpoint 75000, the PSNR of Repeated propagation (Our method) is 0.2 point greater than Coupled and slightly lower in NIQE score. Having 2 over 3 times better in the score, we concluded the comparison between Repeated and Coupled propagation with Repeated propagation of our. Our Repeated propagation is the best suitable for the next steps.

4.1.2 Alignment

In VSR, spatial alignment is crucial because it straightens out strongly related but mis-aligned pictures and characteristics for later aggregation. In the alignment part, we adopt the alignment on features with optical flow approach. Because of the idea of "repeated" and "recalled" above, we introduce a concatenation of pre-aligned features between warping and refine step. The aligned features are then passed to multiple residual blocks for features refinement and correction. In the mathematics notations, we have:

$$\begin{aligned}
 s_i^{b,f} &= S(x_i, x_{i\pm 1}), \\
 w_i^{b,f} &= W(pf_i, s_i^{b,f}), \\
 pf_i^{b,f} &= R^{b,f}(RS(w_i^{b,f}, rf_i, pf_{i\pm 1})) \\
 rf_i &= \text{Concat}(rf_i, pf_i)
 \end{aligned} \tag{4.5}$$

Here RS, S and W denote the features recall summation, flow estimation using SPyNet [23] and spatial warping modules on feature level, respectively, and $R^{b,f}$ denotes a stack of residual convolution blocks for refinement.

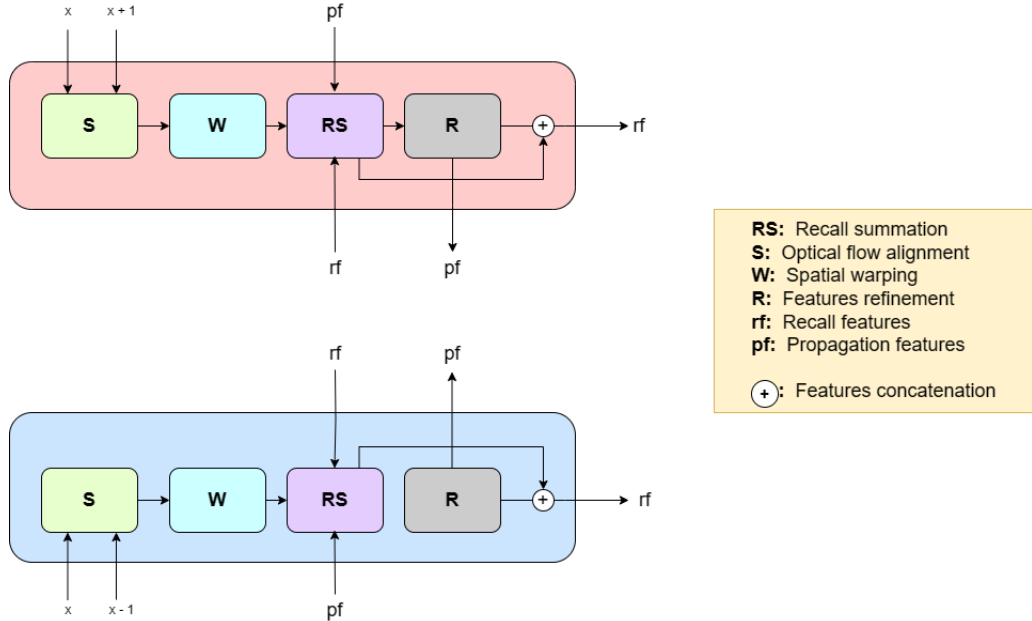


Figure 4.8: Backward and Forward alignment

4.1.3 Aggregation and Upsampling

Our model adopts basic components for aggregation and upsampling. Specifically, given the intermediate features rf_i , to create the final HR image, an upsampling module made up of several convolutions and pixel shuffle is used:

$$y_i = U(rf_i) \quad (4.6)$$

where U denotes the upsampling module, the aggregation is performed by features concatenation as input for convolutions in upsampling module.

4.2 Extend for Real-world application

Recently, majority of the VSR models are trained under the assumption that the LR video are downsampled from the ground truth HR images with a fixed known kernel, such as bicubic. However, real-world LR video frames contain various different types of degradations, including spatial downsampling and motion blurs. One of the proposed solution to solve this problem is by learning as much as possible. In the other words, we need to collect enough training data for all kinds of degradations, which is computationally infeasible. Another solution can be recommend as building a module [26] that can accurately estimate unknown downsample kernel for each video. Unfortunately, these methods require training the estimation network from scratch at inference time and typically runs in minutes for a frame. In previous work, RealBasicVSR [20] introduce a module that instead of predict the unknown kernel, it will transform the unknown kernel to a general kernel (interpolated area) then applying upsampling operations on that transformed kernel. And it's called *cleaning module*.

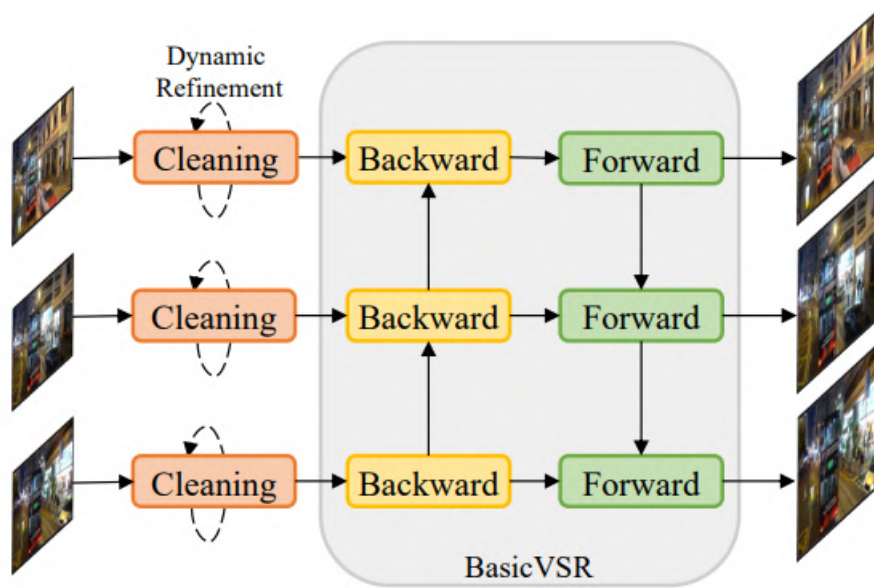


Figure 4.9: RealBasicVSR architecture

4.2.1 Cleaning module formula

The cleaning module on VSR apply the cleaning operation frame by frame. Although it is said to be cleaning module but actually its task is transforming the blind LR to another non-blind LR (Area interpolation) with same low-features structure. The input frames are first independently passed to the cleaning module for degradation transformation. Let I_i^{LR} be the i -th LR frame of the input sequence, and $Clean$ be the image cleaning module, the operation can be described as:

$$(\tilde{I}_i^{LR})^1 = Clean((I_i^{LR})^0) \quad (4.7)$$

The clean sequence then passed to the VSR network SR for Super-Resolution:

$$\hat{y}_i = SR(\tilde{I}_i^{LR}) \quad (4.8)$$

To guide the image cleaning module, RealBasicVSR constrains the outputs of the cleaning module with a low-resolution ground-truth:

$$\begin{aligned} L_{clean} &= mean(L) = \{l_1, l_2, \dots, l_N\}^T \\ l_n &= \|\tilde{I}_i^{LR} - d(y_i)\|_1 \end{aligned} \quad (4.9)$$

Where y_i is the ground-truth high-resolution image, and d is a downsampling operator, here we use area interpolation for downsampling [27]. In addition to the cleaning loss, RealBasicVSR also uses the output fidelity loss to guide the cleaning module.

4.2.2 Cleaning schedule

A single pass of input to the cleaning module cannot effectively remove the excessive degradations in many challenging cases. A simple yet effective method is to suppress the degradations further with another pass to the cleaning module. Formally, RealBasicVSR designs a refinement scheme that dynamically removes the degradations in test time:

$$\begin{cases} (\tilde{I}_i^{LR})^{j+1} = Clean((I_i^{LR})^j) & \text{if } mean(Clean((I_i^{LR})^j)) \geq \theta \\ \tilde{I}_i^{LR} = (\tilde{I}_i^{LR})^j & \text{otherwise,} \end{cases} \quad (4.10)$$

θ is a pre-determined stopping criteria.

4.2.3 Experiment with Cleaning module

Observation: Cleaning module eliminates the artifacts as well as valuable textures.

Cleaning module is used to eliminate undesirable artifacts or noise while maintaining valuable textures or features by clearly inspecting it as well as its behavior since it is the initial input into the model of RealBasicVSR. However, it is critical to strike a balance between removing undesirable elements and retaining essential information. Important details may be lost or changed if the cleaning procedure is overly harsh, resulting in a drop in overall quality. Knowing this, the creators of RealBasicVSR chose to dynamically clean the input three times in order to decrease artifacts as much as possible.

As we already mentioned about the real-world visual representation, there are many kind of degradations. Which means that, there are video sequence that can be harshly degraded, but, the existence of simple degraded videos cannot be denied. This leads to the fact that the assumption of transforming many different blind LRs with unknown kind of degradations into the non-blind LRs degradation by Area interpolation then, fell into the case of over-smoothing and over-cleaning in some areas in some LR frames. Particularly, not only cleaning module of RealBasicVSR eliminates the artifacts, but also does it clear the valuable textures, asking the later BasicVSR module to become an implicit generative model to fill into the zero textures that cleaning module has created.



Figure 4.10: REDS-00-00- after cleaning



Figure 4.11: REDS-00-00- before cleaning

**Figure 4.12:** REDS-15-00- after cleaning**Figure 4.13:** REDS-15-00- before cleaning

Folder	000	015
PSNR	34.744	37.752
SSIM	0.939	0.977

Table 4.1: PSNR, SSIM comparison of after and before cleaning image [Link]

As we observed the cleaning output and the input, while there are artifacts that should be eliminated in input low-quality frames, there are valuable textures that low-quality frames contains that later can be cleared by the cleaning module. The cleaned low-quality frames, on the other hand, cleared the image textures and high frequency details but retain the hard structure. The structure similarity (SSIM) shows in the table 4.1 sheds light on the fact that there is only a slightly change in the structure of the image while the Peak signal-to-noise ratio (PSNR) relating the frequency details of the two image, drops from infinite to around 35 for the example of REDS validation folder 000, first frames.

The problem of losing textures can be easily solved by replacing the cleaning module with an image denoising, deblurring or restoration model, however, this change breaks our scope of Computational Efficiency and breaks the goal of light-weight model. This leads to the idea of our new feature extraction scheme for real-world visual representation clearly formulated below.

Feature extraction for real-world visual representation

We propose a feature extraction method for real-world visual representation base on the observation above. The low-quality input frames and the cleaned frames all bring their advantages, what we are trying is to fully utilize the strength of each one.

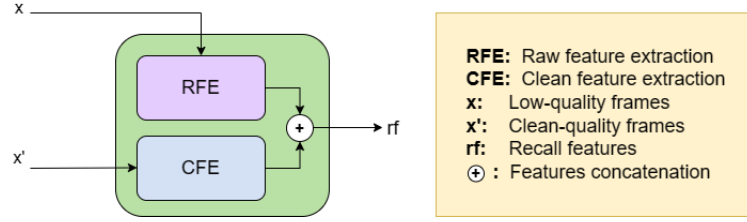


Figure 4.14: Our feature extraction

Specially, given a low-quality frame x , we obtain the cleaned frame x' with the module cleaning consists of multiple convolution layers, then, feed into the Feature extraction module:

$$\begin{aligned} x' &= \text{Clean}(x) \\ rf &= \text{Concat}(\text{RFE}(x), \text{CFE}(x')) \end{aligned} \tag{4.11}$$

We want the RFE to extract the features related to high frequency details which are the textures that can be eliminate by the cleaning module. The CFE is aimed to extract the features of structural textures forming the image, representing the low frequency detail. Combined together with the simple concatenation that permit differentiable flow, we obtain the first recalled features rf to feed into the propagation branches.



Figure 4.15: Input low-quality image



Figure 4.16: RFE feature map sample



Figure 4.17: RFE feature map sample



Figure 4.18: CFE feature map sample



Figure 4.19: CFE feature map sample

Chapter 5

Experiments and Results

5.1 Dataset

As introduced about the evaluation metrics 2.3, it is necessary to collect two types of datasets: Blind dataset (dataset with the unknown degradation kernels) and Non-Blind dataset (dataset with the known degradation kernels).

5.1.1 Non-Blind dataset

Non-Blind dataset implies the dataset that we know the degradation kernels which is applied during the downsample process. . And the downsampling process may just contain one or two basic degradation kernels. In this thesis, we downloaded the evaluation dataset of REDS dataset, which contains two folder, namely: *val_sharp* and *val_sharp_bicubic*. while *val_sharp* folder is the ground truth (720x1280), *val_sharp_bicubic* is the bicubic downsampled version by factor of 4. Both folders contain 30 subfolders (or 30 short videos). Each video is presented as 100 consecutive frames. The illustration of each video can be shown as following:

Furthermore, we also deploy a function named: *paired_random_crop* function, which helps us to crop a paired patch from the corresponding downsampled version and its ground truth for later evaluation. The crop size has been fixed by a reasonable size. In fact, we randomly crop from a low-resolution frame a patch size 64x64 then find the relative location of this patch on the corresponding high-resolution frame to crop (the high-resolution patch

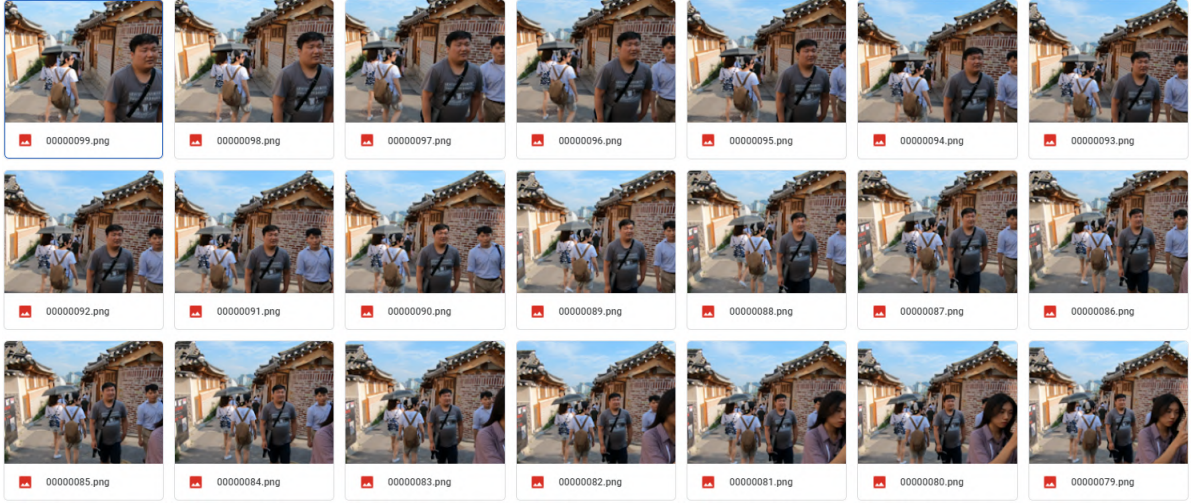


Figure 5.1: REDs evaluation video representation

has a size 256x256). The "crop function" helps us to focus on a specific patch of the frame. And we set the function randomly since we want the evaluation to be general as many as possible. The crop_ dataset can be found by this link .

5.1.2 Blind dataset

Blind dataset implies the dataset that we do not know the degradation kernels which is applied during the downsampling process. The blind dataset contains four 4 folders (4 videos), namely: test1, test2, test3, and test4. Each video has at least 150 consecutive frames. Moreover, the number of motions will significantly increase from test1 to test4. The blind dataset can be found by this link .

5.2 Experiment Settings

5.2.1 Training Degradations

Following Real-ESRGAN [28] and RealBasicVSR [20], we use the multi-stage degradation model and image-based degradations such as random blur, resize, noise, and JPEG compression for the sequence of input frames. In addition, at the video-based degradations we use video compression, which is a popular method for reducing video size using keyframes. Video compression, unlike the aforementioned degradations, implicitly examines the interdependencies between video frames, resulting in temporally and spatially varying degradations. During training, we use compression with randomly chosen codecs and bitrates and notice performance gains with video compression included. We use openly available code of OpenMMLab [29] for general second-order degradation scheme and later for general comparison.

5.2.2 Training Settings

We use the Adam optimizer [30], with the learning rates remain constant of 10^{-4} . Input LR frames will have a patch size of 64×64 . We use the RealBasicVSR stochastic degradation scheme with a temporal sequence of 15 frames and flip the temporal dimension to 30 frames. We first pre-train our model with only output loss and image cleaning loss for 100K iterations, with batch size 2 to reduce the memory size taken by GPU while also accelerate the training procedure; this allows us to fit our model on Google Colab with a Tesla T4 16GB GPU.

After that, we fine-tune our model follow the idea of Real-ESRGAN with the concept of generator (RealBasicVSR) and discriminator of Real-ESRGAN to make our model more robust to the real-world video in next 100k iteration. In this training phase, due to the limit of GPU training pre-defined on phase 1, the discriminator is also allocated for memory, we reduce the batch size to 1 during the training phase. This helps us to remain the training resources while also receiving benefit for training time consideration.

5.3 Results

In this section, we conduct two general types of evaluation: Quantitative and Qualitative evaluation. Quantitative evaluation deals with numbers and statistics, while qualitative evaluation deals with human intuition. Both are important for gaining different kinds of knowledge.

We acknowledged 3 models namely BasicVSR, Real-ESRGAN and RealBasicVSR to make the comparison of 2 aforementioned evaluation strategies.

For BasicVSR and Real-ESRGAN, we use the pre-trained model as these 2 models give the relevant, BasicVSR focuses on the video task and Real-ESRGAN focuses on the Real-world Super-Resolution task, while thankfully eliminating the computational problem of re-training these 2 models. For RealBasicVSR, as this model we base heavily on and also share the same aim to solve the problem of Real-world Video Super-Resolution, we retrained it with the same configs to make the general assumption on the Computation Efficiency scope, the release checkpoint is also include in comparison to have clearer view of training resources different.

5.3.1 Quantitative evaluation

In this part, we compare the 4 models including our together with a basic math method of Bicubic.

We use NIQE score as a metric to justify the naturalness. This metric will be use to compare for all the models and Bicubic method. The NIQE score is calculated with the dataset of VideoLQ [20], which has 50 folders cover various scene of Real-world captured into video type.

	Bicubic	BasicVSR	Real-ESRGAN	RealBasicVSR Retrain	Our
Params (M)	-	6.3	16.7	6.3	6.4
NIQE [17] ↓	8.0049	8.8386	4.2091	4.0579	4.0086

Table 5.1: No Reference Quantitative Comparison. Runtime is computed with an output size of 720×1280, with a Tesla T4 GPU. Red and blue colors represent the best and second best performance, respectively

Furthermore, the Real-world Super-resolution models rarely consider on the structure and high frequency detail of the image, however, it’s really important that we should obtain both the structural preservation and naturalness preservation. While PSNR focuses on objective fidelity, SSIM adds a perceptual dimension to the assessment, making them complementary metrics in the field of video quality evaluation. With the reason of that, PSNR and SSIM is employed for the comparison of our model with only other Real-world task related one namely: Real-ESRGAN, RealBasicVSR. Both score is computed on REDS4 dataset as it provides low quality together with ground truth image.

	Real-ESRGAN	RealBasicVSR Retrain	Our
Params (M)	16.7	6.3	6.4
PSNR ↑	23.85	24.09	24.39
SSIM ↑	0.6702	0.6686	0.6868

Table 5.2: Reference Quantitative Comparison. Runtime is computed with an output size of 720×1280, with a Tesla T4 GPU. Red and blue colors represent the best and second best performance, respectively

5.3.2 Qualitative evaluation

We show examples on 2 folders of REDS4 dataset. Equipped with the image cleaning module and feature extraction baseline, our model is able to aggregate long-term information through propagation effectively while retaining the key important textures. As a result, it generates much more details in fine regions, improving visual quality. For instance, only our model is able to recover the textures of the shirt and hair in the example of folder 011 of REDS4, when coming to the folder 000, our model performance can be stated to be the best, having closest visual quality with the ground truth.

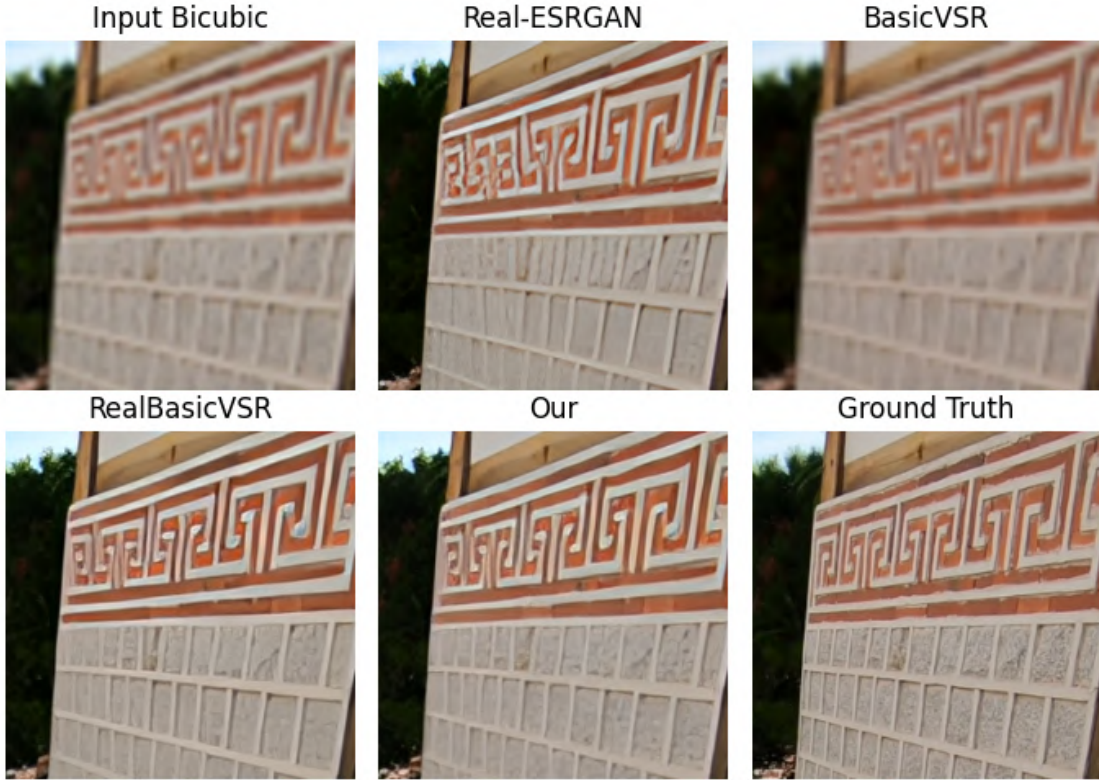


Figure 5.2: Qualitative evaluation 1. Our model image inference on REDS4 folder 000. The texture of our model is well-preserved compared with other models. The color is more accurate with ground truth.

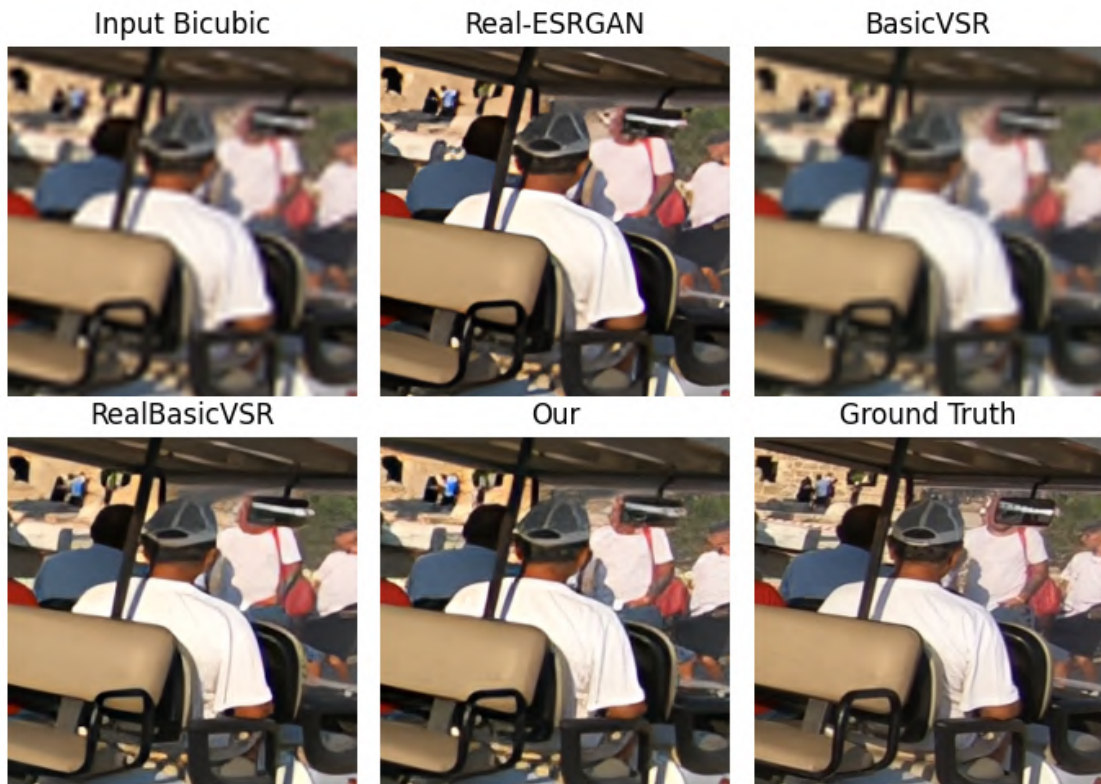


Figure 5.3: Qualitative evaluation 2. Our model image inference on REDS4 folder 011. Only our model, the shirt and hair details are restored.

Chapter 6

Summary and Future Plan

Video Super Resolution is a difficult subject to examine in depth because it is the next step up from Single Image Super Resolution. Considering the ongoing environment of a video and how to provide improved quality makes the execution of the solution more globally relevant as technological devices such as smartphones with enormous storage and tremendous camera while our human demand for entertainment expand over time. As the continuous cannot be turned into the discrete status of video, the degrading of a particular aspect cannot be predicted, hence making our selection susceptible to shortcomings. To inspect a new baseline of a new model, setting up the storage and training GPU appears to be a demanding task. There have been models that run into the case where the allocation is insufficient. We have conducted surveys and experiments to realize our model with small number of parameters and training steps so that it can fit the scarcity of training resources. Having done our utmost, the project has now essentially accomplished the tasks to realize our best model and results verification.

In the next steps, having more time to dive deep into our baseline, we will develop a method to optimize model and realize a more robust method to support the task of Video Super-Resolution. We stated that the only problem we encounter is the training resource problem, the generalization of batch size turns out to be very important, having reduce the batch size of our model so much compared to all of the Computer Vision method for Super Resolution gives the fact that our model fell into local minima of the training set and can not turn back to find a better minima. This will also be our consideration, to find a strong resources for training, testing and achieving new results.

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